

CATS<sup>+</sup>

Competishun Advance

Training Sessions

(Lecture - X)



Ques

Evaluate  $\int_1^9 \left( \frac{1}{2} + \sqrt{\frac{1}{4} + \log_3 x} \right) dx + \int_1^2 \frac{3^{x^2 - [x]}}{3^{\{x\}}} dx$

$$\begin{aligned} &= \int_1^9 \frac{1}{2} \cdot dx + \int_1^9 \sqrt{\frac{1}{4} + \log_3(x)} \cdot dx \\ &= \int_1^2 \frac{x^2 - [x] - \{x\}}{3^{x^2 - [x] - \{x\}}} dx \\ &= \int_1^2 \frac{x^2 - x}{3^{x^2 - x}} dx \end{aligned}$$

$= I_2$

4. ✓

$\sqrt{\frac{1}{4} + \log_3(x)} = t$

$\log_3(x) = t^2 - \frac{1}{4}$

$x = 3^{t^2 - \frac{1}{4}}$

$dx = 3^{t^2 - \frac{1}{4}} \cdot \ln(3) \cdot 2t \cdot dt$

$x=1 \rightarrow t=\frac{1}{2}$

$x=9 \rightarrow t=\frac{3}{2}$



$$I_1 = \int_1^9 \sqrt{\frac{1}{4} + \log_3(x)} \, dx = \int_{\text{I}}^{\text{II}} t \cdot \underbrace{3^{t^2 - \frac{1}{4}} \ln(3) \cdot 2t \cdot dt}_{\text{II}}$$

$$= \left[ t \cdot 3^{t^2 - \frac{1}{4}} \right]_{\frac{1}{2}}^{3/2} - \int_{\frac{1}{2}}^{3/2} 1 \cdot 3^{t^2 - \frac{1}{4}} \, dt$$

$$= \left( \frac{3}{2} \cdot 3^2 - \frac{1}{2} \cdot 1 \right) - \int_{\frac{1}{2}}^{3/2} 3^{t^2 - \frac{1}{4}} \, dt$$

$$\frac{27}{2} - \frac{1}{2} = 13$$

$$\text{R.I.} = 4 + I_1 + I_2 = 4 + 13 - \cancel{I_2} + \cancel{I_2}$$

$$= \boxed{17} \text{ Ans}$$

$$I_2 = \int_1^2 3^{x^2 - x} \, dx$$

$$= \int_1^2 3^{(x - \frac{1}{2})^2 - \frac{1}{4}} \, dx$$

$$= \int_{\frac{1}{2}}^{3/2} 3^{y^2 - \frac{1}{4}} \, dy$$

$$\begin{aligned} x - \frac{1}{2} &= y \\ dx &= dy \end{aligned}$$



Ques

Two tangents  $x - 2y + 10 = 0$  and  $x = 2$  to a parabola intersect the tangent at vertex at  $(-2, 4)$  and  $(2, 0)$  respectively then which of the following is/are true.

- ☒ (A) If co-ordinates of focus is  $(\alpha, \beta)$  then  $\alpha + \beta = 2$
- ☒ (B) If equation of directrix is expressed in the form of  $ax + by = 4$  then  $a + b = 2$ .
- ☒ (C) length of latus rectum is  $4\sqrt{2}$
- ☒ (D) If co-ordinates of vertex is  $(\alpha, \beta)$  then  $\alpha + \beta = 2$ .

Sol

$$T_1 \equiv x - 2y + 10 = 0 \rightarrow A: (-2, 4)$$

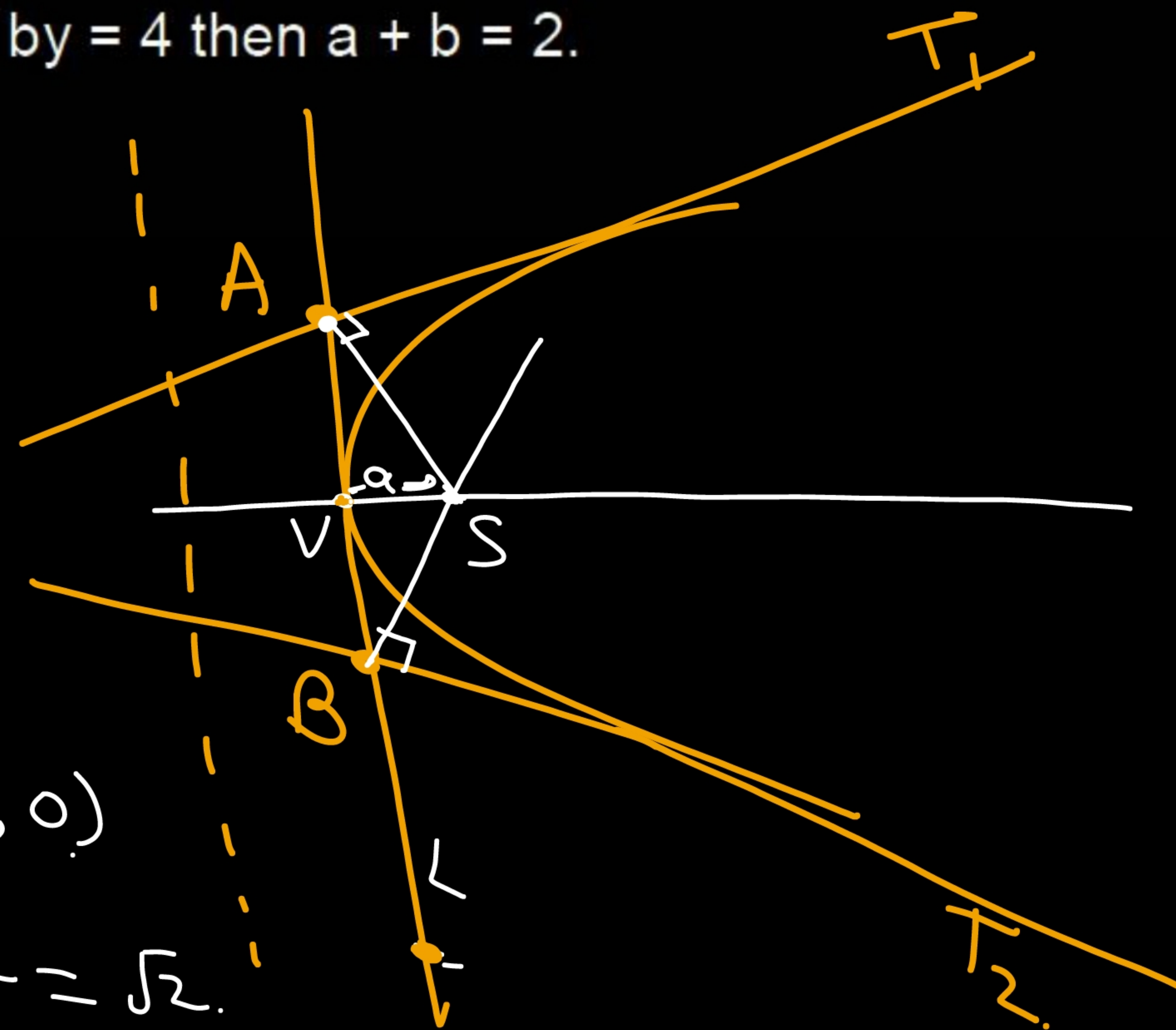
$$T_2 \equiv x = 2 \rightarrow B: (2, 0)$$

$$AS \equiv y - 4 = -2(x + 2) \Rightarrow y = -2x$$

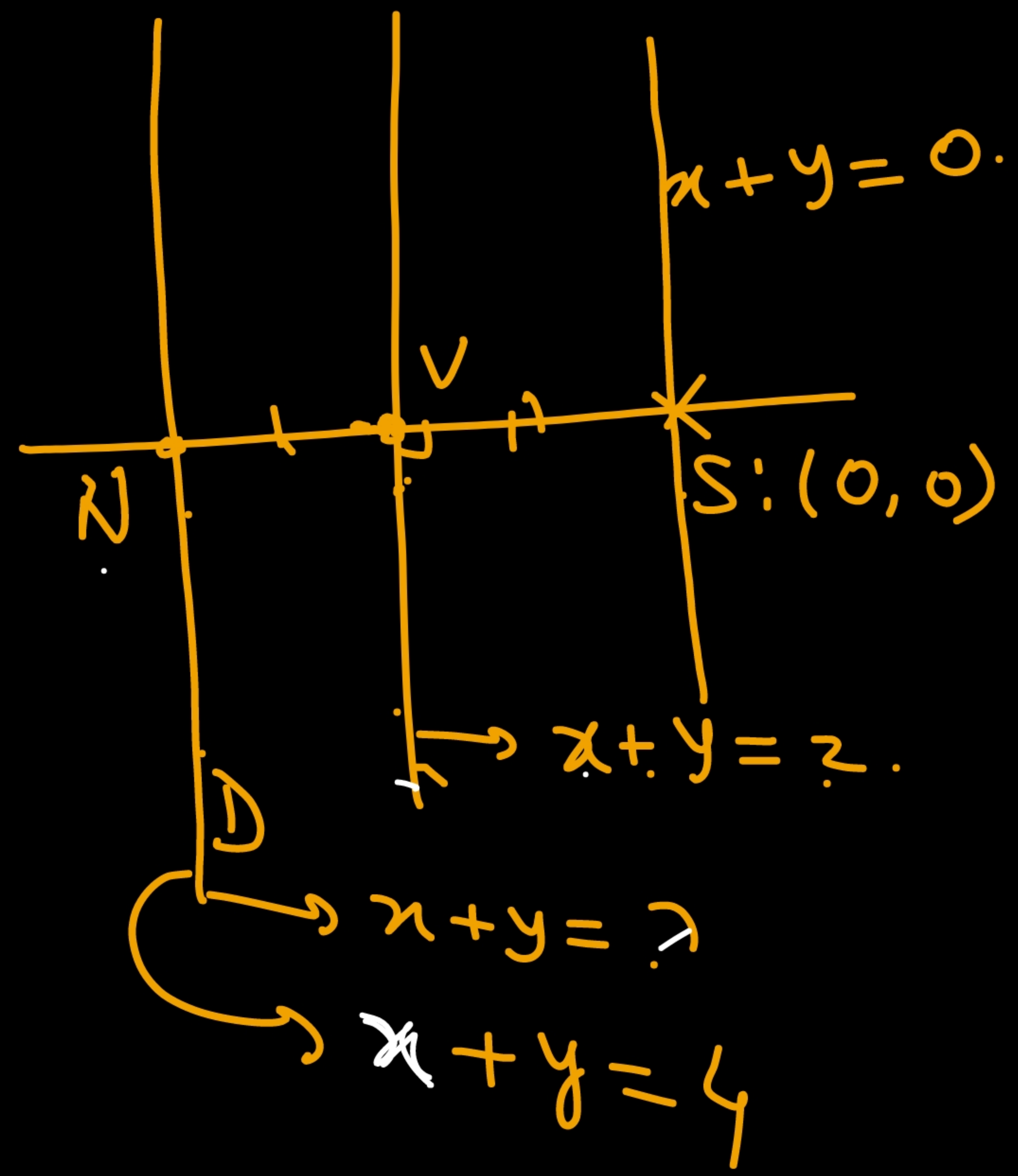
$$BS \equiv y = 0 \Rightarrow S: (0, 0)$$

$$L \equiv y - 0 = \frac{4 - 0}{-2 - 2}(x - 2)$$
$$y = -x + 2 \Rightarrow \boxed{x + y = 2}$$

$$a = \frac{2}{\sqrt{2}} = \sqrt{2}$$
$$L.R. = 4\sqrt{2}$$









Que

The integral part of  $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{10000}}$  is

(A) 196

(B) 197

(C) 198

(D) 200

Sol

$$\begin{aligned} & \sqrt{n-1} < \sqrt{n} < \sqrt{n+1} \\ & + \sqrt{n} \quad \left[ \begin{aligned} & \sqrt{n} + \sqrt{n-1} < 2\sqrt{n} < \sqrt{n} + \sqrt{n+1} \end{aligned} \right. \end{aligned}$$

$$\Rightarrow \left[ \begin{aligned} & \frac{1}{\sqrt{n} - \sqrt{n-1}} > \frac{1}{2\sqrt{n}} > \frac{1}{\sqrt{n} + \sqrt{n+1}} \end{aligned} \right.$$

$$\Rightarrow \frac{2}{\sqrt{n} + \sqrt{n+1}} < \frac{1}{\sqrt{n}} < \frac{2}{\sqrt{n} - \sqrt{n-1}} \Rightarrow 2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$



$$2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$

$n = 2, 3, 4, \dots, 10^4$

$$2 \left( \cancel{(\sqrt{2} - \sqrt{1})} + \cancel{(\sqrt{3} - \sqrt{2})} + \cancel{(\sqrt{4} - \sqrt{3})} + \dots + (\sqrt{10^4 + 1} - \cancel{\sqrt{10^4}}) \right) < \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{10^4}} < 2 \left( \cancel{(\sqrt{2} - \sqrt{1})} + \cancel{(\sqrt{3} - \sqrt{2})} + \cancel{(\sqrt{4} - \sqrt{3})} + \dots + (\sqrt{10^4} - \cancel{\sqrt{10^4 - 1}}) \right)$$

$$2(\sqrt{10^4 + 1} - \sqrt{2}) < R. Sum < 2(\sqrt{10^4} - 1)$$

$\sqrt{10^4 + 1} \approx 100.00001$   
 $\sqrt{2} \approx 1.414$   
 $2(\sqrt{10^4 + 1} - \sqrt{2}) \approx 198.6$   
 $2(\sqrt{10^4} - 1) = 198$

$$\begin{array}{r} 198.6 \\ - 197.2 \\ \hline 1.4 \end{array}$$

$\Rightarrow$

$$197 < R. Sum < 198$$





Que

Find the number of ordered pair  $(p, q)$ , where  $p, q \in \mathbb{N}$  if the equation  $(x^2 - px + q)(x^2 - qx + p) = 0$  has all roots as natural numbers

3

$$p, q \in \mathbb{N}.$$

$$Q_1 \equiv x^2 - px + q = 0 \rightarrow \alpha, \beta$$

$$Q_2 \equiv x^2 - qx + p = 0 \rightarrow \gamma, \delta$$

acc. to question  $\alpha, \beta, \gamma, \delta \in \mathbb{N}$ .

$$D_1 \geq 0 \Rightarrow p^2 - 4q \geq 0$$

$$p^2 \geq 4q$$

$$D_2 \geq 0 \Rightarrow q^2 \geq 4p$$

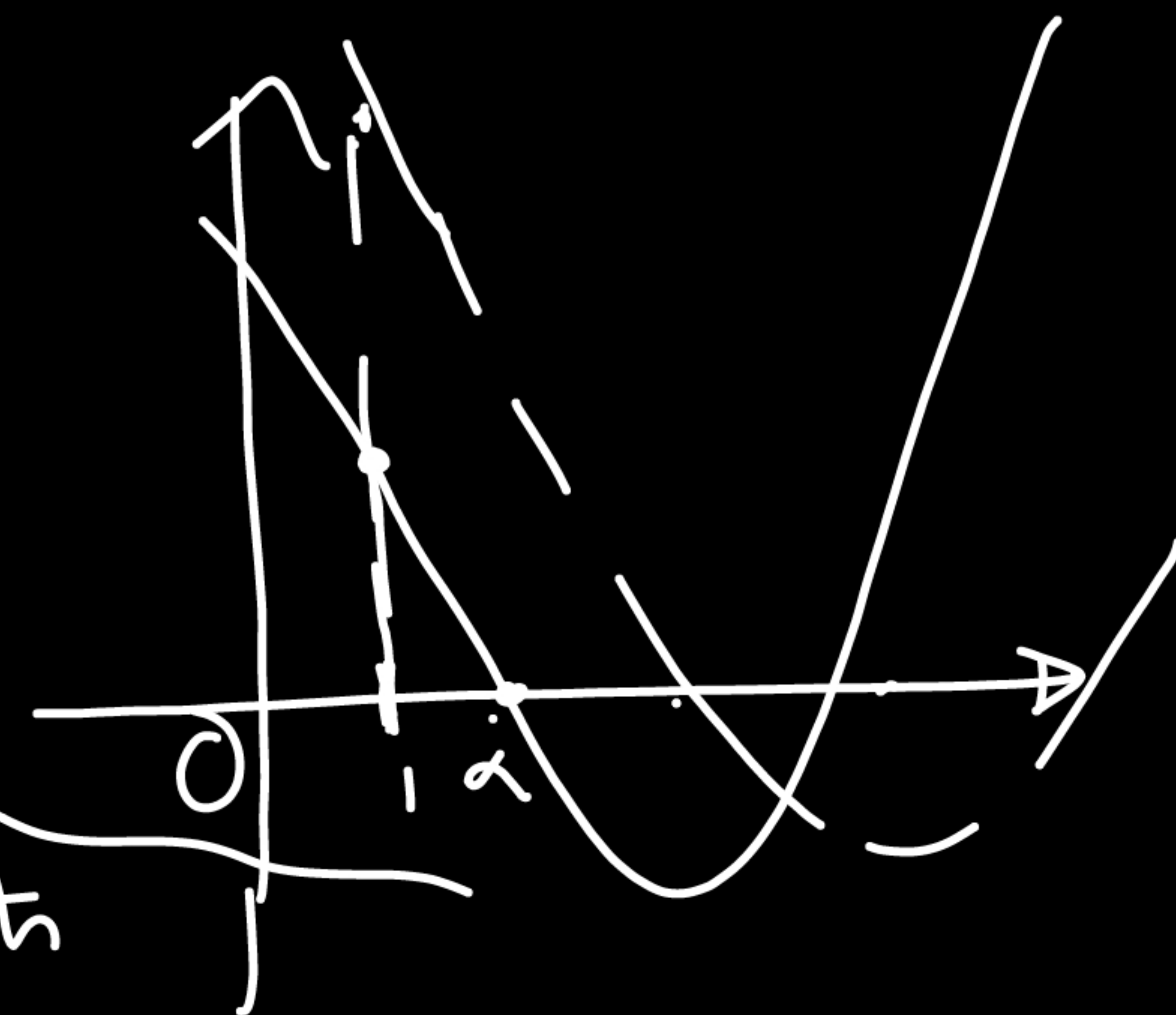
$$\Rightarrow q^4 \geq 16p^2 \geq 16 \cdot 4q$$

$$q^3 \geq 4^3 \Rightarrow q \geq 4$$

$$p \geq 4 \quad \textcircled{8} \quad q \geq 4.$$

$$p^2 \geq 4q$$

$$q^2 \geq 4p$$



if least of the roots is  $> 1$

$$Q_1(1) > 0 \Rightarrow 1 - p + q > 0$$

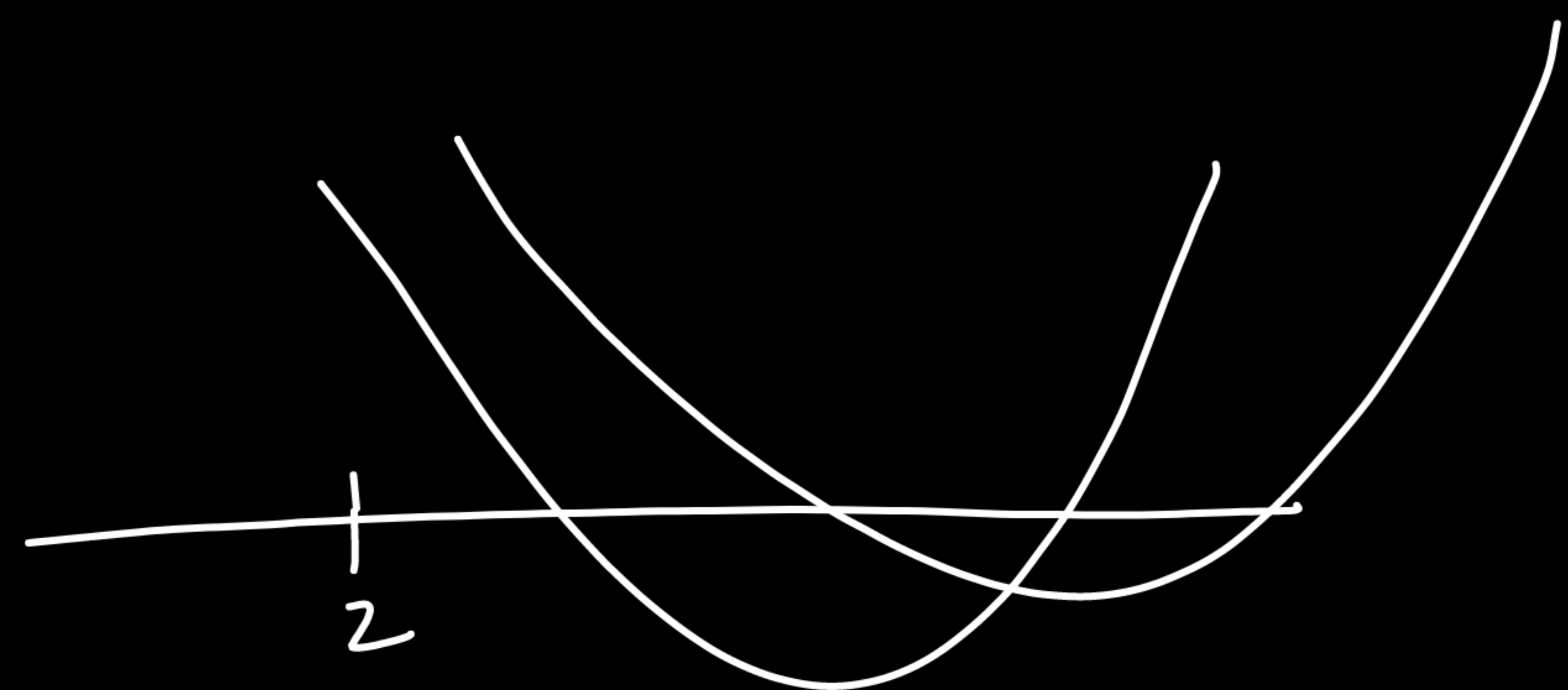
$$Q_2(1) > 0 \Rightarrow 1 - q + p > 0$$

$$\pm \quad 2 > 0 \checkmark$$

Similarly  
 $p \geq 4$



If least of the roots is  $> 2$ .



$$Q_1(2) > 0 \Rightarrow 4 - 2p + q > 0$$

$$Q_2(2) > 0 \Rightarrow 4 - 2q + p > 0$$

$$\begin{array}{r} + \\ \hline 8 - (p+q) > 0 \end{array}$$

$$Q_1 = x^2 - px + q$$

$$\Rightarrow \boxed{p+q < 8}$$

$\Rightarrow$  Not possible

$$x^2 - px + q = 0 \rightarrow \alpha, \gamma$$

$$x^2 - qx + p = 0 \rightarrow \beta, \delta$$

$$1 - p + q = 0$$

$$p = 1 + q$$

$$\hookrightarrow x^2 - qx + (1+q) = 0$$

$$D = q^2 - 4(1+q) = I_1^2$$

$$q^2 - 4q - 4 = I_1^2$$

$$(q-2)^2 - I_1^2 = 8$$

$$(q-2-I_1)(q-2+I_1) = 8$$

$\times 0$

$\{$

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$$0 \leq q-2-I_1+2I_1$$

C-1

$2, \alpha$

$\beta, \gamma$

$$(q-2-I_1)(q-2+I_1) = 8$$

$$\leq 2 \quad \leq 4$$

$$q-2-I_1 = 2$$

$$q-2+I_1 = 4$$

$$q = 5$$



C-1

$$\begin{cases} x^2 - px + q = 1, 5 \\ \Rightarrow x^2 - 6x + 5 = 0. \end{cases} \hookrightarrow 1, 5$$

$$\begin{cases} x^2 - rx + p = 0. \\ \Rightarrow x^2 - 5x + 6 = 0 \end{cases} \hookrightarrow 2, 3$$

C-2.

$$\begin{aligned} x^2 - px + q &= 0 \longrightarrow 2, \alpha \\ x=2 \begin{cases} \Rightarrow 4 - 2p + q = 0. \\ \Rightarrow q = 2(p-2) \end{cases} \end{aligned}$$

$$x^2 - qx + p = 0 \longleftarrow$$

$$x^2 - 2(p-2)x + p = 0.$$

$$D = 4(p-2)^2 - 4p = (2I)^2$$

$$\begin{cases} (p-2)^2 - p = I^2 \end{cases}$$

$$\Rightarrow p^2 - 5p + 4 = I^2 \Rightarrow p^2 - 5p + 4 - I^2 = 0.$$

$$\begin{aligned} D_N &= 25 - 4(4 - I^2) = I_1^2. \\ &= 9 + 4I^2 = I_1^2 \end{aligned}$$



$$9 + 4I^2 = I_1^2$$

$$(I_1 - 2I)(I_1 + 2I) = 9$$

$$\begin{matrix} \leq 1 \\ 3 \end{matrix} \quad \begin{matrix} \leq 9 \\ 3 \end{matrix} \longrightarrow \underline{I} = 0$$

$$\begin{matrix} I_1 - 2I = 1 \\ I_1 + 2I = 9 \end{matrix} \xrightarrow{+} \begin{matrix} 4I = 8 \\ I = 2 \end{matrix}$$

$$p^2 - 5p + 4 = \underline{I^2} = 4$$

$$p^2 - 5p = 0$$

$$\Rightarrow p = 5$$

$$\underline{x^2 - px + q = 0 \rightarrow 2, 3}$$

Ans

$$p^2 - 5p + 4 = 0$$

$$p = 1, 4$$

$$x^2 - px + q = 0 \rightarrow 2, \alpha$$

$$2 + \alpha = p = 4 \Rightarrow \alpha = 2$$

$$\rightarrow x^2 - 4x + 4 = 0$$

$$x^2 - 2x + p = 0$$



|                             | <u>p-1</u>                               | <u>p-2</u>                               |
|-----------------------------|------------------------------------------|------------------------------------------|
| $x^2 - px + q = 0$          | 2, 2                                     | 1, 5                                     |
| $x^2 - \cancel{p}x + p = 0$ | 2, 2                                     | <u><u>2, 3</u></u>                       |
|                             | $p = q = 4$                              | ↓                                        |
|                             | $\begin{matrix} p=5 \\ q=6 \end{matrix}$ | $\begin{matrix} p=6 \\ q=5 \end{matrix}$ |



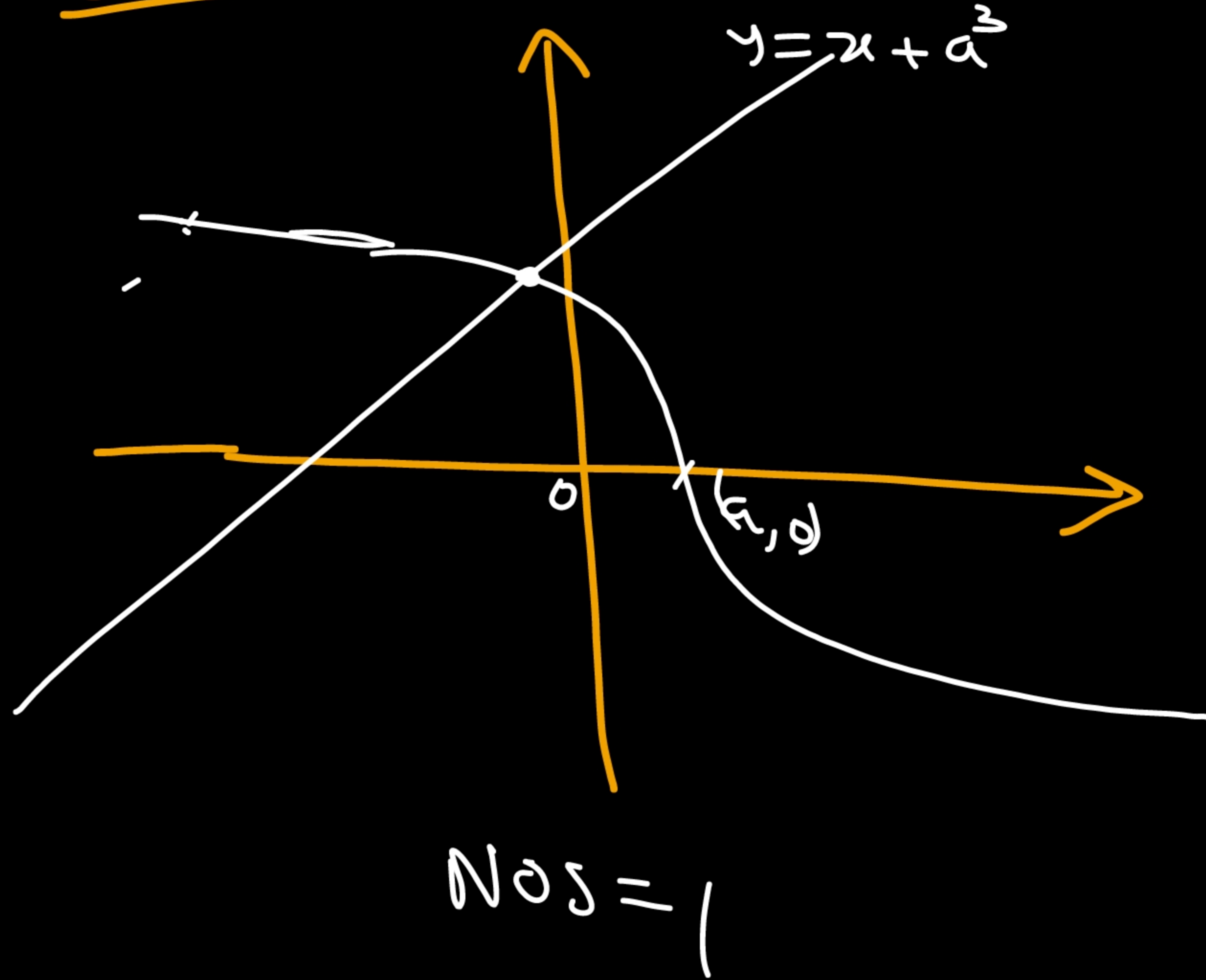
Que The number of solutions of equation  $x + a^3 = (a - x)^{1/3}$  for  $a \in \mathbb{R}$  is  
 (A) 0 (B) 1 (C) 2 (D) 3

Sol

$$x + a^3 = (a - x)^{1/3}, \quad a \in \mathbb{R}.$$

$$\leq (- (x - a))^{1/3}$$

C-1. If  $a > 0$  then



C-2  $a = 0$

$$x = (-x)^{1/3}$$

$Nos = 1$

C-3  $a < 0$

